

Legendre Polynomials

May 7, 2025

Legendre Polynomials and Spherical Harmonics

The Legendre Polynomials have applications to Spherical Harmonic Interpolation. Recall the function $L(f) = \lambda f$. Solutions are given in terms of spherical harmonics:

$$Y_n^m(\theta, \phi) = (-1)^m \sqrt{\frac{2n+1}{4\pi} \cdot \frac{(n-m)!}{(n+m)!}} P_n^m(\cos \theta) e^{im\phi}$$

Gaussian Quadrature Approximation

They are also applicable to numerical integration techniques like the Gaussian Quadrature. Numerical integration provides an approximation of integrals of complex functions. For example, the 2-point Gauss-Legendre rule uses the roots of the second order Legendre Polynomial $\phi_2(x)$ to generate a numerical equation that precisely computes the integral of cubic polynomials.

$$\phi_2(x) = x^2 - \frac{1}{3} = 0$$

$$x = \pm \frac{1}{\sqrt{3}}$$

$$I_n(f_3) = f_3\left(\frac{-1}{\sqrt{3}}\right) + f_3\left(\frac{1}{\sqrt{3}}\right)$$

Gauss-Legendre rule using the second order Legendre Polynomial

$$\phi_2(x) = x^2 - \frac{1}{3} = 0 \quad \text{and its roots } x = \pm \frac{1}{\sqrt{3}}$$

$$I_n(f_3) = f_3\left(-\frac{1}{\sqrt{3}}\right) + f_3\left(\frac{1}{\sqrt{3}}\right)$$

Defining Legendre Polynomials

- Legendre Polynomials are a complete system of orthogonal polynomials over the interval $[-1, 1]$
- They can be defined in a number of ways. List a few below:
 - orthogonal system
 - generating functions with power series
 - differential equation

Rodrigues' Formula:

Rodrigues' formula provides an explicit and closed-form expression for computing Legendre polynomials. It's especially useful for deriving properties like orthogonality and generating functions.

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

Compute the first few low order Legendre Polynomials based on Rodrigues' Formula:

$$P_0(x) = \underline{1}$$

$$P_1(x) = \frac{1}{2} \frac{d}{dx} (x^2 - 1) = x$$

$$P_2(x) = \frac{1}{2} \frac{d^2}{dx^2} (x^2 - 1)^2 = \frac{1}{2} \frac{d}{dx} (4x(x^2 - 1)) = \frac{1}{2} (12x^2 - 4)$$

$$P_3(x) = \frac{1}{2^3 3!} \frac{d^3}{dx^3} (x^2 - 1)^3$$

Bonnet's Recursion:

Derived from Rodrigues' Formula, Bonnet's Recursion is a practical method to generate Polynomials for numerical computations.

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$$

Legendre Polynomial Characteristics

For functions, evenness/Oddness and symmetry help us to analyze the function's behavior. In our case, Legendre Polynomials can be characterized based on their index:

- If n is even: $P_n(x)$ is even, since $P_n(-x) = P_n(x)$
- If n is odd: $P_n(x)$ is odd, since $P_n(-x) = -P_n(x)$

Orthogonality

Orthogonality:

Recall the general definition of orthogonality. We say that two functions, f and g , are orthogonal if:

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x)dx = 0$$

For Legendre Polynomials:

$$\langle P_n, P_m \rangle = \int_{-1}^1 P_n(x)P_m(x)dx = 0 \quad \text{where } m \neq n$$

Similarly, in the Euclidean space:

$$\vec{u} \cdot \vec{v} = 0 \quad \text{and } \vec{u} \neq \vec{v}$$